

# Rolling a 24-word seed — the method

This sheet explains how to use the BIP-39 word table on the pages after the worksheet, for a **24-word seed**. The full procedure, including what to do with the words once you have them, is at [bitcoinkeys.guide/roll-your-own-seed](https://bitcoinkeys.guide/roll-your-own-seed) — read it before you begin. Do not work from this sheet alone.

## What you need

One six-sided die, one coin, pen and paper. No computer, no phone.

## How many words you roll

**23 words.** Your seed is 24 words long, and you roll all but the last one. **You do not roll word 24.** It carries a checksum that has to be calculated, so there is no entry for it in the table and no way to reach it with a die — your hardware wallet works it out, and the procedure on the site explains how for each device.

## For each of your 23 words

1. **Throw the die 5 times. Reroll every 5 and 6** — they do not count and are not written down. Keep throwing until you have 5 results that are each 1, 2, 3 or 4. This is why no entry in the BIP-39 word table contains a 5 or a 6.
2. **Flip the coin once.** H is heads, T is tails. The coin is not optional: the die gives ten of the eleven bits a word needs and the coin gives the eleventh. Always taking the same side halves the randomness of every word you pick.
3. **Find your word.** Blocks are headed by your first 3 throws. Find that block **in the BIP-39 word table**, then read down to the line matching your remaining 2 throws and your flip.
4. **Write the word down, in order, by hand.** Never type it, photograph it, or read it aloud in earshot of anything with a microphone.

## Two things worth knowing

**The BIP-39 word table holds no secret.** It is the standard word list in a fixed order, and anyone can regenerate it.

Printing it, photographing it or keeping it on a phone is harmless. Your *throws* and the *words* they produce are the secret, and those never touch a screen.

**This is the same method and the same table BitBox publishes** for its own devices. We generated ours independently from the word list and checked all 2,048 entries against theirs: every one matches. Either table gives the same word for the same throws — but use one table for a whole seed, not a mix.

# Worksheet — 24-word seed

The moment you write on this sheet it becomes your seed. Treat it exactly as you would the words themselves: never photograph it, never type it up, never leave it out. When your words are safely on their permanent backup and you have restored from that backup successfully, destroy this sheet — shred or burn it. Do not simply bin it.

| WORD # | BLOCK (1st 3 rolls)                          |  |  | LINE (next 2) |  | FLIP (H/T) | WORD (BIP-39 table) |
|--------|----------------------------------------------|--|--|---------------|--|------------|---------------------|
|        |                                              |  |  |               |  |            | <i>globe</i>        |
| 1      |                                              |  |  |               |  |            |                     |
| 2      |                                              |  |  |               |  |            |                     |
| 3      |                                              |  |  |               |  |            |                     |
| 4      |                                              |  |  |               |  |            |                     |
| 5      |                                              |  |  |               |  |            |                     |
| 6      |                                              |  |  |               |  |            |                     |
| 7      |                                              |  |  |               |  |            |                     |
| 8      |                                              |  |  |               |  |            |                     |
| 9      |                                              |  |  |               |  |            |                     |
| 10     |                                              |  |  |               |  |            |                     |
| 11     |                                              |  |  |               |  |            |                     |
| 12     |                                              |  |  |               |  |            |                     |
| 13     |                                              |  |  |               |  |            |                     |
| 14     |                                              |  |  |               |  |            |                     |
| 15     |                                              |  |  |               |  |            |                     |
| 16     |                                              |  |  |               |  |            |                     |
| 17     |                                              |  |  |               |  |            |                     |
| 18     |                                              |  |  |               |  |            |                     |
| 19     |                                              |  |  |               |  |            |                     |
| 20     |                                              |  |  |               |  |            |                     |
| 21     |                                              |  |  |               |  |            |                     |
| 22     |                                              |  |  |               |  |            |                     |
| 23     |                                              |  |  |               |  |            |                     |
| 24     | Not rolled — your device calculates this one |  |  |               |  |            |                     |

This sheet is for a 24-word seed: you roll 23 words and your device supplies word 24, which carries the checksum. A 12-word seed has its own sheet — print only the one you are doing. Full procedure: [bitcoinkeys.guide/roll-your-own-seed](https://bitcoinkeys.guide/roll-your-own-seed)

## 11

## 111

|              |              |
|--------------|--------------|
| 11H abandon  | 31H acoustic |
| 11T ability  | 31T acquire  |
| 12H able     | 32H across   |
| 12T about    | 32T act      |
| 13H above    | 33H action   |
| 13T absent   | 33T actor    |
| 14H absorb   | 34H actress  |
| 14T abstract | 34T actual   |
| 21H absurd   | 41H adapt    |
| 21T abuse    | 41T add      |
| 22H access   | 42H addict   |
| 22T accident | 42T address  |
| 23H account  | 43H adjust   |
| 23T accuse   | 43T admit    |
| 24H achieve  | 44H adult    |
| 24T acid     | 44T advance  |

## 112

|             |             |
|-------------|-------------|
| 11H advice  | 31H alcohol |
| 11T aerobic | 31T alert   |
| 12H affair  | 32H alien   |
| 12T afford  | 32T all     |
| 13H afraid  | 33H alley   |
| 13T again   | 33T allow   |
| 14H age     | 34H almost  |
| 14T agent   | 34T alone   |
| 21H agree   | 41H alpha   |
| 21T ahead   | 41T already |
| 22H aim     | 42H also    |
| 22T air     | 42T alter   |
| 23H airport | 43H always  |
| 23T aisle   | 43T amateur |
| 24H alarm   | 44H amazing |
| 24T album   | 44T among   |

## 113

|              |             |
|--------------|-------------|
| 11H amount   | 31H anxiety |
| 11T amused   | 31T any     |
| 12H analyst  | 32H apart   |
| 12T anchor   | 32T apology |
| 13H ancient  | 33H appear  |
| 13T anger    | 33T apple   |
| 14H angle    | 34H approve |
| 14T angry    | 34T april   |
| 21H animal   | 41H arch    |
| 21T ankle    | 41T arctic  |
| 22H announce | 42H area    |
| 22T annual   | 42T arena   |
| 23H another  | 43H argue   |
| 23T answer   | 43T arm     |
| 24H antenna  | 44H armed   |
| 24T antique  | 44T armor   |

## 114

|              |              |
|--------------|--------------|
| 11H army     | 31H asthma   |
| 11T around   | 31T athlete  |
| 12H arrange  | 32H atom     |
| 12T arrest   | 32T attack   |
| 13H arrive   | 33H attend   |
| 13T arrow    | 33T attitude |
| 14H art      | 34H attract  |
| 14T artefact | 34T auction  |
| 21H artist   | 41H audit    |
| 21T artwork  | 41T august   |
| 22H ask      | 42H aunt     |
| 22T aspect   | 42T author   |
| 23H assault  | 43H auto     |
| 23T asset    | 43T autumn   |
| 24H assist   | 44H average  |
| 24T assume   | 44T avocado  |

## 12

## 121

|              |             |
|--------------|-------------|
| 11H avoid    | 31H bamboo  |
| 11T awake    | 31T banana  |
| 12H aware    | 32H banner  |
| 12T away     | 32T bar     |
| 13H awesome  | 33H barely  |
| 13T awful    | 33T bargain |
| 14H awkward  | 34H barrel  |
| 14T axis     | 34T base    |
| 21H baby     | 41H basic   |
| 21T bachelor | 41T basket  |
| 22H bacon    | 42H battle  |
| 22T badge    | 42T beach   |
| 23H bag      | 43H bean    |
| 23T balance  | 43T beauty  |
| 24H balcony  | 44H because |
| 24T ball     | 44T become  |

## 122

|             |             |
|-------------|-------------|
| 11H beef    | 31H bid     |
| 11T before  | 31T bike    |
| 12H begin   | 32H bind    |
| 12T behave  | 32T biology |
| 13H behind  | 33H bird    |
| 13T believe | 33T birth   |
| 14H below   | 34H bitter  |
| 14T belt    | 34T black   |
| 21H bench   | 41H blade   |
| 21T benefit | 41T blame   |
| 22H best    | 42H blanket |
| 22T betray  | 42T blast   |
| 23H better  | 43H bleak   |
| 23T between | 43T bless   |
| 24H beyond  | 44H blind   |
| 24T bicycle | 44T blood   |

## 123

|             |             |
|-------------|-------------|
| 11H blossom | 31H borrow  |
| 11T blouse  | 31T boss    |
| 12H blue    | 32H bottom  |
| 12T blur    | 32T bounce  |
| 13H blush   | 33H box     |
| 13T board   | 33T boy     |
| 14H boat    | 34H bracket |
| 14T body    | 34T brain   |
| 21H boil    | 41H brand   |
| 21T bomb    | 41T brass   |
| 22H bone    | 42H brave   |
| 22T bonus   | 42T bread   |
| 23H book    | 43H breeze  |
| 23T boost   | 43T brick   |
| 24H border  | 44H bridge  |
| 24T boring  | 44T brief   |

## 124

|              |              |
|--------------|--------------|
| 11H bright   | 31H bulk     |
| 11T bring    | 31T bullet   |
| 12H brisk    | 32H bundle   |
| 12T broccoli | 32T bunker   |
| 13H broken   | 33H burden   |
| 13T bronze   | 33T burger   |
| 14H broom    | 34H burst    |
| 14T brother  | 34T bus      |
| 21H brown    | 41H business |
| 21T brush    | 41T busy     |
| 22H bubble   | 42H butter   |
| 22T buddy    | 42T buyer    |
| 23H budget   | 43H buzz     |
| 23T buffalo  | 43T cabbage  |
| 24H build    | 44H cabin    |
| 24T bulb     | 44T cable    |

## 13

## 131

|             |             |
|-------------|-------------|
| 11H cactus  | 31H capital |
| 11T cage    | 31T captain |
| 12H cake    | 32H car     |
| 12T call    | 32T carbon  |
| 13H calm    | 33H card    |
| 13T camera  | 33T cargo   |
| 14H camp    | 34H carpet  |
| 14T can     | 34T carry   |
| 21H canal   | 41H cart    |
| 21T cancel  | 41T case    |
| 22H candy   | 42H cash    |
| 22T cannon  | 42T casino  |
| 23H canoe   | 43H castle  |
| 23T canvas  | 43T casual  |
| 24H canyon  | 44H cat     |
| 24T capable | 44T catalog |

## 132

|              |              |
|--------------|--------------|
| 11H catch    | 31H champion |
| 11T category | 31T change   |
| 12H cattle   | 32H chaos    |
| 12T caught   | 32T chapter  |
| 13H cause    | 33H charge   |
| 13T caution  | 33T chase    |
| 14H cave     | 34H chat     |
| 14T ceiling  | 34T cheap    |
| 21H celery   | 41H check    |
| 21T cement   | 41T cheese   |
| 22H census   | 42H chef     |
| 22T century  | 42T cherry   |
| 23H cereal   | 43H chest    |
| 23T certain  | 43T chicken  |
| 24H chair    | 44H chief    |
| 24T chalk    | 44T child    |

## 133

|              |            |
|--------------|------------|
| 11H chimney  | 31H claw   |
| 11T choice   | 31T clay   |
| 12H choose   | 32H clean  |
| 12T chronic  | 32T clerk  |
| 13H chuckle  | 33H clever |
| 13T chunk    | 33T click  |
| 14H churn    | 34H client |
| 14T cigar    | 34T cliff  |
| 21H cinnamon | 41H climb  |
| 21T circle   | 41T clinic |
| 22H citizen  | 42H clip   |
| 22T city     | 42T clock  |
| 23H civil    | 43H clog   |
| 23T claim    | 43T close  |
| 24H clap     | 44H cloth  |
| 24T clarify  | 44T cloud  |

## 134

|             |              |
|-------------|--------------|
| 11H clown   | 31H come     |
| 11T club    | 31T comfort  |
| 12H clump   | 32H comic    |
| 12T cluster | 32T common   |
| 13H clutch  | 33H company  |
| 13T coach   | 33T concert  |
| 14H coast   | 34H conduct  |
| 14T coconut | 34T confirm  |
| 21H code    | 41H congress |
| 21T coffee  | 41T connect  |
| 22H coil    | 42H consider |
| 22T coin    | 42T control  |
| 23H collect | 43H convince |
| 23T color   | 43T cook     |
| 24H column  | 44H cool     |
| 24T combine | 44T copper   |

## 14

## 141

|             |             |
|-------------|-------------|
| 11H copy    | 31H craft   |
| 11T coral   | 31T cram    |
| 12H core    | 32H crane   |
| 12T corn    | 32T crash   |
| 13H correct | 33H crater  |
| 13T cost    | 33T crawl   |
| 14H cotton  | 34H crazy   |
| 14T couch   | 34T cream   |
| 21H country | 41H credit  |
| 21T couple  | 41T creek   |
| 22H course  | 42H crew    |
| 22T cousin  | 42T cricket |
| 23H cover   | 43H crime   |
| 23T coyote  | 43T crisp   |
| 24H crack   | 44H critic  |
| 24T cradle  | 44T crop    |

## 142

|              |              |
|--------------|--------------|
| 11H cross    | 31H current  |
| 11T crouch   | 31T curtain  |
| 12H crowd    | 32H curve    |
| 12T crucial  | 32T cushion  |
| 13H cruel    | 33H custom   |
| 13T cruise   | 33T cute     |
| 14H crumble  | 34H cycle    |
| 14T crunch   | 34T dad      |
| 21H crush    | 41H damage   |
| 21T cry      | 41T damp     |
| 22H crystal  | 42H dance    |
| 22T cube     | 42T danger   |
| 23H culture  | 43H daring   |
| 23T cup      | 43T dash     |
| 24H cupboard | 44H daughter |
| 24T curious  | 44T dawn     |

## 143

|              |              |
|--------------|--------------|
| 11H day      | 31H deliver  |
| 11T deal     | 31T demand   |
| 12H debate   | 32H demise   |
| 12T debris   | 32T denial   |
| 13H decade   | 33H dentist  |
| 13T december | 33T deny     |
| 14H decide   | 34H depart   |
| 14T decline  | 34T depend   |
| 21H decorate | 41H deposit  |
| 21T decrease | 41T depth    |
| 22H deer     | 42H deputy   |
| 22T defense  | 42T derive   |
| 23H define   | 43H describe |
| 23T defy     | 43T desert   |
| 24H degree   | 44H design   |
| 24T delay    | 44T desk     |

## 144

|             |              |
|-------------|--------------|
| 11H despair | 31H dignity  |
| 11T destroy | 31T dilemma  |
| 12H detail  | 32H dinner   |
| 12T detect  | 32T dinosaur |
| 13H develop | 33H direct   |
| 13T device  | 33T dirt     |
| 14H devote  | 34H disagree |
| 14T diagram | 34T discover |
| 21H dial    | 41H disease  |
| 21T diamond | 41T dish     |
| 22H diary   | 42H dismiss  |
| 22T dice    | 42T disorder |
| 23H diesel  | 43H display  |
| 23T diet    | 43T distance |
| 24H differ  | 44H divert   |
| 24T digital | 44T divide   |

## 21

### 211

|              |             |
|--------------|-------------|
| 11H divorce  | 31H dragon  |
| 11T dizzy    | 31T drama   |
| 12H doctor   | 32H drastic |
| 12T document | 32T draw    |
| 13H dog      | 33H dream   |
| 13T doll     | 33T dress   |
| 14H dolphin  | 34H drift   |
| 14T domain   | 34T drill   |
| 21H donate   | 41H drink   |
| 21T donkey   | 41T drip    |
| 22H donor    | 42H drive   |
| 22T door     | 42T drop    |
| 23H dose     | 43H drum    |
| 23T double   | 43T dry     |
| 24H dove     | 44H duck    |
| 24T draft    | 44T dumb    |

### 212

|             |              |
|-------------|--------------|
| 11H dune    | 31H ecology  |
| 11T during  | 31T economy  |
| 12H dust    | 32H edge     |
| 12T dutch   | 32T edit     |
| 13H duty    | 33H educate  |
| 13T dwarf   | 33T effort   |
| 14H dynamic | 34H egg      |
| 14T eager   | 34T eight    |
| 21H eagle   | 41H either   |
| 21T early   | 41T elbow    |
| 22H earn    | 42H elder    |
| 22T earth   | 42T electric |
| 23H easily  | 43H elegant  |
| 23T east    | 43T element  |
| 24H easy    | 44H elephant |
| 24T echo    | 44T elevator |

### 213

|             |              |
|-------------|--------------|
| 11H elite   | 31H energy   |
| 11T else    | 31T enforce  |
| 12H embark  | 32H engage   |
| 12T embody  | 32T engine   |
| 13H embrace | 33H enhance  |
| 13T emerge  | 33T enjoy    |
| 14H emotion | 34H enlist   |
| 14T employ  | 34T enough   |
| 21H empower | 41H enrich   |
| 21T empty   | 41T enroll   |
| 22H enable  | 42H ensure   |
| 22T enact   | 42T enter    |
| 23H end     | 43H entire   |
| 23T endless | 43T entry    |
| 24H endorse | 44H envelope |
| 24T enemy   | 44T episode  |

### 214

|              |              |
|--------------|--------------|
| 11H equal    | 31H evoke    |
| 11T equip    | 31T evolve   |
| 12H era      | 32H exact    |
| 12T erase    | 32T example  |
| 13H erode    | 33H excess   |
| 13T erosion  | 33T exchange |
| 14H error    | 34H excite   |
| 14T erupt    | 34T exclude  |
| 21H escape   | 41H excuse   |
| 21T essay    | 41T execute  |
| 22H essence  | 42H exercise |
| 22T estate   | 42T exhaust  |
| 23H eternal  | 43H exhibit  |
| 23T ethics   | 43T exile    |
| 24H evidence | 44H exist    |
| 24T evil     | 44T exit     |

## 22

### 221

|             |             |
|-------------|-------------|
| 11H exotic  | 31H faith   |
| 11T expand  | 31T fall    |
| 12H expect  | 32H false   |
| 12T expire  | 32T fame    |
| 13H explain | 33H family  |
| 13T expose  | 33T famous  |
| 14H express | 34H fan     |
| 14T extend  | 34T fancy   |
| 21H extra   | 41H fantasy |
| 21T eye     | 41T farm    |
| 22H eyebrow | 42H fashion |
| 22T fabric  | 42T fat     |
| 23H face    | 43H fatal   |
| 23T faculty | 43T father  |
| 24H fade    | 44H fatigue |
| 24T faint   | 44T fault   |

### 222

|              |             |
|--------------|-------------|
| 11H favorite | 31H figure  |
| 11T feature  | 31T file    |
| 12H february | 32H film    |
| 12T federal  | 32T filter  |
| 13H fee      | 33H final   |
| 13T feed     | 33T find    |
| 14H feel     | 34H fine    |
| 14T female   | 34T finger  |
| 21H fence    | 41H finish  |
| 21T festival | 41T fire    |
| 22H fetch    | 42H firm    |
| 22T fever    | 42T first   |
| 23H few      | 43H fiscal  |
| 23T fiber    | 43T fish    |
| 24H fiction  | 44H fit     |
| 24T field    | 44T fitness |

### 223

|            |             |
|------------|-------------|
| 11H fix    | 31H foam    |
| 11T flag   | 31T focus   |
| 12H flame  | 32H fog     |
| 12T flash  | 32T foil    |
| 13H flat   | 33H fold    |
| 13T flavor | 33T follow  |
| 14H flee   | 34H food    |
| 14T flight | 34T foot    |
| 21H flip   | 41H force   |
| 21T float  | 41T forest  |
| 22H flock  | 42H forget  |
| 22T floor  | 42T fork    |
| 23H flower | 43H fortune |
| 23T fluid  | 43T forum   |
| 24H flush  | 44H forward |
| 24T fly    | 44T fossil  |

### 224

|              |             |
|--------------|-------------|
| 11H foster   | 31H fun     |
| 11T found    | 31T funny   |
| 12H fox      | 32H furnace |
| 12T fragile  | 32T fury    |
| 13H frame    | 33H future  |
| 13T frequent | 33T gadget  |
| 14H fresh    | 34H gain    |
| 14T friend   | 34T galaxy  |
| 21H fringe   | 41H gallery |
| 21T frog     | 41T game    |
| 22H front    | 42H gap     |
| 22T frost    | 42T garage  |
| 23H frown    | 43H garbage |
| 23T frozen   | 43T garden  |
| 24H fruit    | 44H garlic  |
| 24T fuel     | 44T garment |

## 23

### 231

|             |             |
|-------------|-------------|
| 11H gas     | 31H ginger  |
| 11T gasp    | 31T giraffe |
| 12H gate    | 32H girl    |
| 12T gather  | 32T give    |
| 13H gauge   | 33H glad    |
| 13T gaze    | 33T glance  |
| 14H general | 34H glare   |
| 14T genius  | 34T glass   |
| 21H genre   | 41H glide   |
| 21T gentle  | 41T glimpse |
| 22H genuine | 42H globe   |
| 22T gesture | 42T gloom   |
| 23H ghost   | 43H glory   |
| 23T giant   | 43T glove   |
| 24H gift    | 44H glow    |
| 24T giggle  | 44T glue    |

### 232

|             |             |
|-------------|-------------|
| 11H goat    | 31H gravity |
| 11T goddess | 31T great   |
| 12H gold    | 32H green   |
| 12T good    | 32T grid    |
| 13H goose   | 33H grief   |
| 13T gorilla | 33T grit    |
| 14H gospel  | 34H grocery |
| 14T gossip  | 34T group   |
| 21H govern  | 41H grow    |
| 21T gown    | 41T grunt   |
| 22H grab    | 42H guard   |
| 22T grace   | 42T guess   |
| 23H grain   | 43H guide   |
| 23T grant   | 43T guilt   |
| 24H grape   | 44H guitar  |
| 24T grass   | 44T gun     |

### 233

|             |              |
|-------------|--------------|
| 11H gym     | 31H head     |
| 11T habit   | 31T health   |
| 12H hair    | 32H heart    |
| 12T half    | 32T heavy    |
| 13H hammer  | 33H hedgehog |
| 13T hamster | 33T height   |
| 14H hand    | 34H hello    |
| 14T happy   | 34T helmet   |
| 21H harbor  | 41H help     |
| 21T hard    | 41T hen      |
| 22H harsh   | 42H hero     |
| 22T harvest | 42T hidden   |
| 23H hat     | 43H high     |
| 23T have    | 43T hill     |
| 24H hawk    | 44H hint     |
| 24T hazard  | 44T hip      |

### 234

|              |             |
|--------------|-------------|
| 11H hire     | 31H host    |
| 11T history  | 31T hotel   |
| 12H hobby    | 32H hour    |
| 12T hockey   | 32T hover   |
| 13H hold     | 33H hub     |
| 13T hole     | 33T huge    |
| 14H holiday  | 34H human   |
| 14T hollow   | 34T humble  |
| 21H home     | 41H humor   |
| 21T honey    | 41T hundred |
| 22H hood     | 42H hungry  |
| 22T hope     | 42T hunt    |
| 23H horn     | 43H hurdle  |
| 23T horror   | 43T hurry   |
| 24H horse    | 44H hurt    |
| 24T hospital | 44T husband |

## 24

### 241

|              |              |
|--------------|--------------|
| 11H hybrid   | 31H improve  |
| 11T ice      | 31T impulse  |
| 12H icon     | 32H inch     |
| 12T idea     | 32T include  |
| 13H identify | 33H income   |
| 13T idle     | 33T increase |
| 14H ignore   | 34H index    |
| 14T ill      | 34T indicate |
| 21H illegal  | 41H indoor   |
| 21T illness  | 41T industry |
| 22H image    | 42H infant   |
| 22T imitate  | 42T inflict  |
| 23H immense  | 43H inform   |
| 23T immune   | 43T inhale   |
| 24H impact   | 44H inherit  |
| 24T impose   | 44T initial  |

### 242

|              |             |
|--------------|-------------|
| 11H inject   | 31H invite  |
| 11T injury   | 31T involve |
| 12H inmate   | 32H iron    |
| 12T inner    | 32T island  |
| 13H innocent | 33H isolate |
| 13T input    | 33T issue   |
| 14H inquiry  | 34H item    |
| 14T insane   | 34T ivory   |
| 21H insect   | 41H jacket  |
| 21T inside   | 41T jaguar  |
| 22H inspire  | 42H jar     |
| 22T install  | 42T jazz    |
| 23H intact   | 43H jealous |
| 23T interest | 43T jeans   |
| 24H into     | 44H jelly   |
| 24T invest   | 44T jewel   |

### 243

|              |             |
|--------------|-------------|
| 11H job      | 31H key     |
| 11T join     | 31T kick    |
| 12H joke     | 32H kid     |
| 12T journey  | 32T kidney  |
| 13H joy      | 33H kind    |
| 13T judge    | 33T kingdom |
| 14H juice    | 34H kiss    |
| 14T jump     | 34T kit     |
| 21H jungle   | 41H kitchen |
| 21T junior   | 41T kite    |
| 22H junk     | 42H kitten  |
| 22T just     | 42T kiwi    |
| 23H kangaroo | 43H knee    |
| 23T keen     | 43T knife   |
| 24H keep     | 44H knock   |
| 24T ketchup  | 44T know    |

### 244

|              |             |
|--------------|-------------|
| 11H lab      | 31H lawn    |
| 11T label    | 31T lawsuit |
| 12H labor    | 32H layer   |
| 12T ladder   | 32T lazy    |
| 13H lady     | 33H leader  |
| 13T lake     | 33T leaf    |
| 14H lamp     | 34H learn   |
| 14T language | 34T leave   |
| 21H laptop   | 41H lecture |
| 21T large    | 41T left    |
| 22H later    | 42H leg     |
| 22T latin    | 42T legal   |
| 23H laugh    | 43H legend  |
| 23T laundry  | 43T leisure |
| 24H lava     | 44H lemon   |
| 24T law      | 44T lend    |

## 31

### 311

|             |             |
|-------------|-------------|
| 11H length  | 31H link    |
| 11T lens    | 31T lion    |
| 12H leopard | 32H liquid  |
| 12T lesson  | 32T list    |
| 13H letter  | 33H little  |
| 13T level   | 33T live    |
| 14H liar    | 34H lizard  |
| 14T liberty | 34T load    |
| 21H library | 41H loan    |
| 21T license | 41T lobster |
| 22H life    | 42H local   |
| 22T lift    | 42T lock    |
| 23H light   | 43H logic   |
| 23T like    | 43T lonely  |
| 24H limb    | 44H long    |
| 24T limit   | 44T loop    |

### 312

|             |             |
|-------------|-------------|
| 11H lottery | 31H maid    |
| 11T loud    | 31T mail    |
| 12H lounge  | 32H main    |
| 12T love    | 32T major   |
| 13H loyal   | 33H make    |
| 13T lucky   | 33T mammal  |
| 14H luggage | 34H man     |
| 14T lumber  | 34T manage  |
| 21H lunar   | 41H mandate |
| 21T lunch   | 41T mango   |
| 22H luxury  | 42H mansion |
| 22T lyrics  | 42T manual  |
| 23H machine | 43H maple   |
| 23T mad     | 43T marble  |
| 24H magic   | 44H march   |
| 24T magnet  | 44T margin  |

### 313

|              |              |
|--------------|--------------|
| 11H marine   | 31H meat     |
| 11T market   | 31T mechanic |
| 12H marriage | 32H medal    |
| 12T mask     | 32T media    |
| 13H mass     | 33H melody   |
| 13T master   | 33T melt     |
| 14H match    | 34H member   |
| 14T material | 34T memory   |
| 21H math     | 41H mention  |
| 21T matrix   | 41T menu     |
| 22H matter   | 42H mercy    |
| 22T maximum  | 42T merge    |
| 23H maze     | 43H merit    |
| 23T meadow   | 43T merry    |
| 24H mean     | 44H mesh     |
| 24T measure  | 44T message  |

### 314

|              |             |
|--------------|-------------|
| 11H metal    | 31H mix     |
| 11T method   | 31T mixed   |
| 12H middle   | 32H mixture |
| 12T midnight | 32T mobile  |
| 13H milk     | 33H model   |
| 13T million  | 33T modify  |
| 14H mimic    | 34H mom     |
| 14T mind     | 34T moment  |
| 21H minimum  | 41H monitor |
| 21T minor    | 41T monkey  |
| 22H minute   | 42H monster |
| 22T miracle  | 42T month   |
| 23H mirror   | 43H moon    |
| 23T misery   | 43T moral   |
| 24H miss     | 44H more    |
| 24T mistake  | 44T morning |

## 32

### 321

|              |              |
|--------------|--------------|
| 11H mosquito | 31H must     |
| 11T mother   | 31T mutual   |
| 12H motion   | 32H myself   |
| 12T motor    | 32T mystery  |
| 13H mountain | 33H myth     |
| 13T mouse    | 33T naive    |
| 14H move     | 34H name     |
| 14T movie    | 34T napkin   |
| 21H much     | 41H narrow   |
| 21T muffin   | 41T nasty    |
| 22H mule     | 42H nation   |
| 22T multiply | 42T nature   |
| 23H muscle   | 43H near     |
| 23T museum   | 43T neck     |
| 24H mushroom | 44H need     |
| 24T music    | 44T negative |

### 322

|             |             |
|-------------|-------------|
| 11H neglect | 31H noodle  |
| 11T neither | 31T normal  |
| 12H nephew  | 32H north   |
| 12T nerve   | 32T nose    |
| 13H nest    | 33H notable |
| 13T net     | 33T note    |
| 14H network | 34H nothing |
| 14T neutral | 34T notice  |
| 21H never   | 41H novel   |
| 21T news    | 41T now     |
| 22H next    | 42H nuclear |
| 22T nice    | 42T number  |
| 23H night   | 43H nurse   |
| 23T noble   | 43T nut     |
| 24H noise   | 44H oak     |
| 24T nominee | 44T obey    |

### 323

|             |             |
|-------------|-------------|
| 11H object  | 31H old     |
| 11T oblige  | 31T olive   |
| 12H obscure | 32H olympic |
| 12T observe | 32T omit    |
| 13H obtain  | 33H once    |
| 13T obvious | 33T one     |
| 14H occur   | 34H onion   |
| 14T ocean   | 34T online  |
| 21H october | 41H only    |
| 21T odor    | 41T open    |
| 22H off     | 42H opera   |
| 22T offer   | 42T opinion |
| 23H office  | 43H oppose  |
| 23T often   | 43T option  |
| 24H oil     | 44H orange  |
| 24T okay    | 44T orbit   |

### 324

|              |             |
|--------------|-------------|
| 11H orchard  | 31H own     |
| 11T order    | 31T owner   |
| 12H ordinary | 32H oxygen  |
| 12T organ    | 32T oyster  |
| 13H orient   | 33H ozone   |
| 13T original | 33T pact    |
| 14H orphan   | 34H paddle  |
| 14T ostrich  | 34T page    |
| 21H other    | 41H pair    |
| 21T outdoor  | 41T palace  |
| 22H outer    | 42H palm    |
| 22T output   | 42T panda   |
| 23H outside  | 43H panel   |
| 23T oval     | 43T panic   |
| 24H oven     | 44H panther |
| 24T over     | 44T paper   |

## 33

### 331

|             |              |
|-------------|--------------|
| 11H parade  | 31H pear     |
| 11T parent  | 31T peasant  |
| 12H park    | 32H pelican  |
| 12T parrot  | 32T pen      |
| 13H party   | 33H penalty  |
| 13T pass    | 33T pencil   |
| 14H patch   | 34H people   |
| 14T path    | 34T pepper   |
| 21H patient | 41H perfect  |
| 21T patrol  | 41T permit   |
| 22H pattern | 42H person   |
| 22T pause   | 42T pet      |
| 23H pave    | 43H phone    |
| 23T payment | 43T photo    |
| 24H peace   | 44H phrase   |
| 24T peanut  | 44T physical |

### 332

|             |             |
|-------------|-------------|
| 11H piano   | 31H plastic |
| 11T picnic  | 31T plate   |
| 12H picture | 32H play    |
| 12T piece   | 32T please  |
| 13H pig     | 33H pledge  |
| 13T pigeon  | 33T pluck   |
| 14H pill    | 34H plug    |
| 14T pilot   | 34T plunge  |
| 21H pink    | 41H poem    |
| 21T pioneer | 41T poet    |
| 22H pipe    | 42H point   |
| 22T pistol  | 42T polar   |
| 23H pitch   | 43H pole    |
| 23T pizza   | 43T police  |
| 24H place   | 44H pond    |
| 24T planet  | 44T pony    |

### 333

|              |              |
|--------------|--------------|
| 11H pool     | 31H present  |
| 11T popular  | 31T pretty   |
| 12H portion  | 32H prevent  |
| 12T position | 32T price    |
| 13H possible | 33H pride    |
| 13T post     | 33T primary  |
| 14H potato   | 34H print    |
| 14T pottery  | 34T priority |
| 21H poverty  | 41H prison   |
| 21T powder   | 41T private  |
| 22H power    | 42H prize    |
| 22T practice | 42T problem  |
| 23H praise   | 43H process  |
| 23T predict  | 43T produce  |
| 24H prefer   | 44H profit   |
| 24T prepare  | 44T program  |

### 334

|              |              |
|--------------|--------------|
| 11H project  | 31H puppy    |
| 11T promote  | 31T purchase |
| 12H proof    | 32H purity   |
| 12T property | 32T purpose  |
| 13H prosper  | 33H purse    |
| 13T protect  | 33T push     |
| 14H proud    | 34H put      |
| 14T provide  | 34T puzzle   |
| 21H public   | 41H pyramid  |
| 21T pudding  | 41T quality  |
| 22H pull     | 42H quantum  |
| 22T pulp     | 42T quarter  |
| 23H pulse    | 43H question |
| 23T pumpkin  | 43T quick    |
| 24H punch    | 44H quit     |
| 24T pupil    | 44T quiz     |

## 34

### 341

|             |             |
|-------------|-------------|
| 11H quote   | 31H rare    |
| 11T rabbit  | 31T rate    |
| 12H raccoon | 32H rather  |
| 12T race    | 32T raven   |
| 13H rack    | 33H raw     |
| 13T radar   | 33T razor   |
| 14H radio   | 34H ready   |
| 14T rail    | 34T real    |
| 21H rain    | 41H reason  |
| 21T raise   | 41T rebel   |
| 22H rally   | 42H rebuild |
| 22T ramp    | 42T recall  |
| 23H ranch   | 43H receive |
| 23T random  | 43T recipe  |
| 24H range   | 44H record  |
| 24T rapid   | 44T recycle |

### 342

|              |              |
|--------------|--------------|
| 11H reduce   | 31H render   |
| 11T reflect  | 31T renew    |
| 12H reform   | 32H rent     |
| 12T refuse   | 32T reopen   |
| 13H region   | 33H repair   |
| 13T regret   | 33T repeat   |
| 14H regular  | 34H replace  |
| 14T reject   | 34T report   |
| 21H relax    | 41H require  |
| 21T release  | 41T rescue   |
| 22H relief   | 42H resemble |
| 22T rely     | 42T resist   |
| 23H remain   | 43H resource |
| 23T remember | 43T response |
| 24H remind   | 44H result   |
| 24T remove   | 44T retire   |

### 343

|             |             |
|-------------|-------------|
| 11H retreat | 31H ring    |
| 11T return  | 31T riot    |
| 12H reunion | 32H ripple  |
| 12T reveal  | 32T risk    |
| 13H review  | 33H ritual  |
| 13T reward  | 33T rival   |
| 14H rhythm  | 34H river   |
| 14T rib     | 34T road    |
| 21H ribbon  | 41H roast   |
| 21T rice    | 41T robot   |
| 22H rich    | 42H robust  |
| 22T ride    | 42T rocket  |
| 23H ridge   | 43H romance |
| 23T rifle   | 43T roof    |
| 24H right   | 44H rookie  |
| 24T rigid   | 44T room    |

### 344

|             |             |
|-------------|-------------|
| 11H rose    | 31H safe    |
| 11T rotate  | 31T sail    |
| 12H rough   | 32H salad   |
| 12T round   | 32T salmon  |
| 13H route   | 33H salon   |
| 13T royal   | 33T salt    |
| 14H rubber  | 34H salute  |
| 14T rude    | 34T same    |
| 21H rug     | 41H sample  |
| 21T rule    | 41T sand    |
| 22H run     | 42H satisfy |
| 22T runway  | 42T satoshi |
| 23H rural   | 43H sauce   |
| 23T sad     | 43T sausage |
| 24H saddle  | 44H save    |
| 24T sadness | 44T say     |

## 41

### 411

|              |              |
|--------------|--------------|
| 11H scale    | 31H search   |
| 11T scan     | 31T season   |
| 12H scare    | 32H seat     |
| 12T scatter  | 32T second   |
| 13H scene    | 33H secret   |
| 13T scheme   | 33T section  |
| 14H school   | 34H security |
| 14T science  | 34T seed     |
| 21H scissors | 41H seek     |
| 21T scorpion | 41T segment  |
| 22H scout    | 42H select   |
| 22T scrap    | 42T sell     |
| 23H screen   | 43H seminar  |
| 23T script   | 43T senior   |
| 24H scrub    | 44H sense    |
| 24T sea      | 44T sentence |

### 412

|             |              |
|-------------|--------------|
| 11H series  | 31H ship     |
| 11T service | 31T shiver   |
| 12H session | 32H shock    |
| 12T settle  | 32T shoe     |
| 13H setup   | 33H shoot    |
| 13T seven   | 33T shop     |
| 14H shadow  | 34H short    |
| 14T shaft   | 34T shoulder |
| 21H shallow | 41H shove    |
| 21T share   | 41T shrimp   |
| 22H shed    | 42H shrug    |
| 22T shell   | 42T shuffle  |
| 23H sheriff | 43H shy      |
| 23T shield  | 43T sibling  |
| 24H shift   | 44H sick     |
| 24T shine   | 44T side     |

### 413

|             |             |
|-------------|-------------|
| 11H siege   | 31H skate   |
| 11T sight   | 31T sketch  |
| 12H sign    | 32H ski     |
| 12T silent  | 32T skill   |
| 13H silk    | 33H skin    |
| 13T silly   | 33T skirt   |
| 14H silver  | 34H skull   |
| 14T similar | 34T slab    |
| 21H simple  | 41H slam    |
| 21T since   | 41T sleep   |
| 22H sing    | 42H slender |
| 22T siren   | 42T slice   |
| 23H sister  | 43H slide   |
| 23T situate | 43T slight  |
| 24H six     | 44H slim    |
| 24T size    | 44T slogan  |

### 414

|            |              |
|------------|--------------|
| 11H slot   | 31H sock     |
| 11T slow   | 31T soda     |
| 12H slush  | 32H soft     |
| 12T small  | 32T solar    |
| 13H smart  | 33H soldier  |
| 13T smile  | 33T solid    |
| 14H smoke  | 34H solution |
| 14T smooth | 34T solve    |
| 21H snack  | 41H someone  |
| 21T snake  | 41T song     |
| 22H snap   | 42H soon     |
| 22T sniff  | 42T sorry    |
| 23H snow   | 43H sort     |
| 23T soap   | 43T soul     |
| 24H soccer | 44H sound    |
| 24T social | 44T soup     |

## 42

### 421

|             |              |
|-------------|--------------|
| 11H source  | 31H spirit   |
| 11T south   | 31T split    |
| 12H space   | 32H spoil    |
| 12T spare   | 32T sponsor  |
| 13H spatial | 33H spoon    |
| 13T spawn   | 33T sport    |
| 14H speak   | 34H spot     |
| 14T special | 34T spray    |
| 21H speed   | 41H spread   |
| 21T spell   | 41T spring   |
| 22H spend   | 42H spy      |
| 22T sphere  | 42T square   |
| 23H spice   | 43H squeeze  |
| 23T spider  | 43T squirrel |
| 24H spike   | 44H stable   |
| 24T spin    | 44T stadium  |

### 422

|            |              |
|------------|--------------|
| 11H staff  | 31H stock    |
| 11T stage  | 31T stomach  |
| 12H stairs | 32H stone    |
| 12T stamp  | 32T stool    |
| 13H stand  | 33H story    |
| 13T start  | 33T stove    |
| 14H state  | 34H strategy |
| 14T stay   | 34T street   |
| 21H steak  | 41H strike   |
| 21T steel  | 41T strong   |
| 22H stem   | 42H struggle |
| 22T step   | 42T student  |
| 23H stereo | 43H stuff    |
| 23T stick  | 43T stumble  |
| 24H still  | 44H style    |
| 24T sting  | 44T subject  |

### 423

|             |              |
|-------------|--------------|
| 11H submit  | 31H sure     |
| 11T subway  | 31T surface  |
| 12H success | 32H surge    |
| 12T such    | 32T surprise |
| 13H sudden  | 33H surround |
| 13T suffer  | 33T survey   |
| 14H sugar   | 34H suspect  |
| 14T suggest | 34T sustain  |
| 21H suit    | 41H swallow  |
| 21T summer  | 41T swamp    |
| 22H sun     | 42H swap     |
| 22T sunny   | 42T swarm    |
| 23H sunset  | 43H swear    |
| 23T super   | 43T sweet    |
| 24H supply  | 44H swift    |
| 24T supreme | 44T swim     |

### 424

|             |            |
|-------------|------------|
| 11H swing   | 31H task   |
| 11T switch  | 31T taste  |
| 12H sword   | 32H tattoo |
| 12T symbol  | 32T taxi   |
| 13H symptom | 33H teach  |
| 13T syrup   | 33T team   |
| 14H system  | 34H tell   |
| 14T table   | 34T ten    |
| 21H tackle  | 41H tenant |
| 21T tag     | 41T tennis |
| 22H tail    | 42H tent   |
| 22T talent  | 42T term   |
| 23H talk    | 43H test   |
| 23T tank    | 43T text   |
| 24H tape    | 44H thank  |
| 24T target  | 44T that   |

## 43

### 431

|             |              |
|-------------|--------------|
| 11H theme   | 31H tilt     |
| 11T then    | 31T timber   |
| 12H theory  | 32H time     |
| 12T there   | 32T tiny     |
| 13H they    | 33H tip      |
| 13T thing   | 33T tired    |
| 14H this    | 34H tissue   |
| 14T thought | 34T title    |
| 21H three   | 41H toast    |
| 21T thrive  | 41T tobacco  |
| 22H throw   | 42H today    |
| 22T thumb   | 42T toddler  |
| 23H thunder | 43H toe      |
| 23T ticket  | 43T together |
| 24H tide    | 44H toilet   |
| 24T tiger   | 44T token    |

### 432

|              |              |
|--------------|--------------|
| 11H tomato   | 31H toward   |
| 11T tomorrow | 31T tower    |
| 12H tone     | 32H town     |
| 12T tongue   | 32T toy      |
| 13H tonight  | 33H track    |
| 13T tool     | 33T trade    |
| 14H tooth    | 34H traffic  |
| 14T top      | 34T tragic   |
| 21H topic    | 41H train    |
| 21T topple   | 41T transfer |
| 22H torch    | 42H trap     |
| 22T tornado  | 42T trash    |
| 23H tortoise | 43H travel   |
| 23T toss     | 43T tray     |
| 24H total    | 44H treat    |
| 24T tourist  | 44T tree     |

### 433

|             |             |
|-------------|-------------|
| 11H trend   | 31H tube    |
| 11T trial   | 31T tuition |
| 12H tribe   | 32H tumble  |
| 12T trick   | 32T tuna    |
| 13H trigger | 33H tunnel  |
| 13T trim    | 33T turkey  |
| 14H trip    | 34H turn    |
| 14T trophy  | 34T turtle  |
| 21H trouble | 41H twelve  |
| 21T truck   | 41T twenty  |
| 22H true    | 42H twice   |
| 22T truly   | 42T twin    |
| 23H trumpet | 43H twist   |
| 23T trust   | 43T two     |
| 24H truth   | 44H type    |
| 24T try     | 44T typical |

### 434

|              |             |
|--------------|-------------|
| 11H ugly     | 31H unlock  |
| 11T umbrella | 31T until   |
| 12H unable   | 32H unusual |
| 12T unaware  | 32T unveil  |
| 13H uncle    | 33H update  |
| 13T uncover  | 33T upgrade |
| 14H under    | 34H uphold  |
| 14T undo     | 34T upon    |
| 21H unfair   | 41H upper   |
| 21T unfold   | 41T upset   |
| 22H unhappy  | 42H urban   |
| 22T uniform  | 42T urge    |
| 23H unique   | 43H usage   |
| 23T unit     | 43T use     |
| 24H universe | 44H used    |
| 24T unknown  | 44T useful  |

## 44

### 441

|             |             |
|-------------|-------------|
| 11H useless | 31H velvet  |
| 11T usual   | 31T vendor  |
| 12H utility | 32H venture |
| 12T vacant  | 32T venue   |
| 13H vacuum  | 33H verb    |
| 13T vague   | 33T verify  |
| 14H valid   | 34H version |
| 14T valley  | 34T very    |
| 21H valve   | 41H vessel  |
| 21T van     | 41T veteran |
| 22H vanish  | 42H viable  |
| 22T vapor   | 42T vibrant |
| 23H various | 43H vicious |
| 23T vast    | 43T victory |
| 24H vault   | 44H video   |
| 24T vehicle | 44T view    |

### 442

|             |             |
|-------------|-------------|
| 11H village | 31H voyage  |
| 11T vintage | 31T wage    |
| 12H violin  | 32H wagon   |
| 12T virtual | 32T wait    |
| 13H virus   | 33H walk    |
| 13T visa    | 33T wall    |
| 14H visit   | 34H walnut  |
| 14T visual  | 34T want    |
| 21H vital   | 41H warfare |
| 21T vivid   | 41T warm    |
| 22H vocal   | 42H warrior |
| 22T voice   | 42T wash    |
| 23H void    | 43H wasp    |
| 23T volcano | 43T waste   |
| 24H volume  | 44H water   |
| 24T vote    | 44T wave    |

### 443

|             |             |
|-------------|-------------|
| 11H way     | 31H wheel   |
| 11T wealth  | 31T when    |
| 12H weapon  | 32H where   |
| 12T wear    | 32T whip    |
| 13H weasel  | 33H whisper |
| 13T weather | 33T wide    |
| 14H web     | 34H width   |
| 14T wedding | 34T wife    |
| 21H weekend | 41H wild    |
| 21T weird   | 41T will    |
| 22H welcome | 42H win     |
| 22T west    | 42T window  |
| 23H wet     | 43H wine    |
| 23T whale   | 43T wing    |
| 24H what    | 44H wink    |
| 24T wheat   | 44T winner  |

### 444

|             |             |
|-------------|-------------|
| 11H winter  | 31H wrap    |
| 11T wire    | 31T wreck   |
| 12H wisdom  | 32H wrestle |
| 12T wise    | 32T wrist   |
| 13H wish    | 33H write   |
| 13T witness | 33T wrong   |
| 14H wolf    | 34H yard    |
| 14T woman   | 34T year    |
| 21H wonder  | 41H yellow  |
| 21T wood    | 41T you     |
| 22H wool    | 42H young   |
| 22T word    | 42T youth   |
| 23H work    | 43H zebra   |
| 23T world   | 43T zero    |
| 24H worry   | 44H zone    |
| 24T worth   | 44T zoo     |